Self-consistent Majorana nanowire models

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I. HARTREE-FOCK-NAMBU MEAN-FIELD THEORY

Consider a generic interacting Hamiltonian $H = H_0 + H_{\rm int}$ of the form

$$\hat{H} = \sum_{\alpha\beta} h_{\alpha\beta} c^\dagger_\alpha c_\beta + \frac{1}{2}\sum_{\alpha\beta\gamma\delta} V^{\alpha\beta}_{\gamma\delta} c^\dagger_\alpha c^\dagger_\beta c_\gamma c_\delta, \qquad (1)$$

with $c^\dagger(c)$ fermionic creation (annihilation) operators and α, β, γ, δ generic degrees of freedom. The time-dependency on h and fermionic operators being omitted for clarity. Concerning the interaction term, we assume $V(\mathbf{r})$ to be a generic electrostatic (i.e spin independent) potential of symmetry $V(\mathbf{r}-\mathbf{r}') = V(\mathbf{r}'-\mathbf{r})$ such that the interaction tensor elements have the symmetries $V^{\alpha\beta}_{\gamma\delta} = V^{\beta\alpha}_{\delta\gamma}$ and $(V^{\alpha\beta}_{\gamma\delta})^* = V^{\gamma\delta}_{\alpha\beta}$.

This system's purely electronic reduced density matrix (rDM) is defined as $\rho^{\rm ee}_{ba} = \langle c^\dagger_a c_b\rangle$. However, if one wishes to study superconducting systems, one must also account for the anomalous rDM. For this, we introduce the Nambu spinor representation with doubling of degrees of freedom $\check{c}_a = (c_a \;\; c^\dagger_a)$. The Nambu representation of the rDM, can then be written as the tensor product of this spinors for each of the spin combinations,

$$\rho_{ab} = \begin{pmatrix} \langle c^\dagger_b c_a\rangle & \langle c_b c_a\rangle \\ \langle c^\dagger_b c^\dagger_a\rangle & \langle c_b c^\dagger_a\rangle \end{pmatrix} = \begin{pmatrix} \rho^{\rm ee}_{ab} & \rho^{\rm eh}_{ab} \\ \rho^{\rm he}_{ab} & \rho^{\rm hh}_{ab} \end{pmatrix}. \qquad (2)$$

These four contributions do, of course, relate to each other as $\rho^{\rm hh}_{ab} = -\rho^{\rm ee}_{ba} + \delta_{ab}$ and $\rho^{\rm he}_{ab} = (\rho^{\rm eh}_{ba})^*$, but also $\rho^{\rm ee/hh}_{ab} = (\rho^{\rm ee/hh}_{ba})^*$ and $\rho^{\rm eh/he}_{ba} = -\rho^{\rm eh/he}_{ab}$ directly from the (equal-time) fermionic anti-commutator properties.

Assume we wish to study the time-evolution of the Nambu rDM. This amounts to solving both the normal and anomalous rDM equation of motions with the fermionic operators evolving accordingly to the Heisenberg equation. Hence, one needs to evaluate the commutators $[H, c^\dagger_b]$ and $[H, c_a]$ which result in four-operator expectation values. To deal with such term we perform a mean-field decoupling where we approximate a two-particle expectation value to a product of two one-particle's. From Wick's theorem we have that, for example for the purely electronic rDM,

$$\langle c^\dagger_\alpha c^\dagger_\beta c_\gamma c_\delta\rangle \approx \rho^{\rm he}_{\beta\alpha}\rho^{\rm eh}_{\delta\gamma} - \rho^{\rm ee}_{\gamma\alpha}\rho^{\rm ee}_{\delta\beta} + \rho^{\rm ee}_{\delta\alpha}\rho^{\rm ee}_{\gamma\beta}. \qquad (3)$$

Defining Hartree, Fock and pairing self-energies (themselves dependent on the rDM) as

$$\begin{matrix} \Sigma^H_{ab} = \sum_{\gamma\delta} V^{\delta a}_{b\gamma}\rho^{\rm ee}_{\gamma\delta} \\ \Sigma^F_{ab} = -\sum_{\gamma\delta} V^{a\delta}_{b\gamma}\rho^{\rm ee}_{\gamma\delta} \end{matrix} \quad \text{and} \quad \Delta_{ab} = \sum_{\gamma\delta} V^{ab}_{\delta\gamma}\rho^{\rm eh}_{\gamma\delta} \qquad (4)$$

(note the symmetry $\Delta_{ab} = -\Delta_{ba}$) allows one to write the Nambu rDM EoM in terms of an effective Bogoliubov-de Gennes (BdG) Hamiltonian as simply as

$$i\hbar\frac{d}{dt}\rho_{ab} = \left[H^{\rm BdG}_{ab}, \rho_{ab}\right] \text{ with } H^{\rm BdG}_{ab} = \begin{pmatrix} h^{\rm HF}_{ab} & \Delta_{ab} \\ \Delta^*_{ba} & -h^{\rm HF}_{ba} \end{pmatrix}$$

with the Hartree-Fock Hamiltonian being

$$h_{HF}(\rho) = h + \mathbf{\Sigma}_H(\rho) + \mathbf{\Sigma}_F(\rho). \qquad (5)$$

Furthermore, consider now the approximation where the orbitals are ultra-localized (known as Wannier orbitals) such that $V^{\alpha\beta}_{\gamma\delta} \approx V(\mathbf{r}_\alpha - \mathbf{r}_\beta)\delta_{\alpha\delta}\delta_{\beta\gamma}$. Within the scope of this approximation the interaction Hamiltonian will simply read as $H_{\rm int} = \frac{1}{2}\sum_{\alpha\beta} V^{\alpha\beta} c^\dagger_\alpha c^\dagger_\beta c_\beta c_\alpha$ with $V^{\alpha\beta} \equiv V(\mathbf{r}_\alpha - \mathbf{r}_\beta)$. Furthermore, for the sake of getting the problem in a tight-binding scheme, we isolate the degrees of freedom into purely spatial and spin indices $a \to i\sigma_i$ such that

$$\boldsymbol{\rho}_{ij} = \begin{pmatrix} \boldsymbol{\rho}^{\rm ee}_{ij} & \boldsymbol{\rho}^{\rm eh}_{ij} \\ \boldsymbol{\rho}^{\rm he}_{ji} & \boldsymbol{\rho}^{\rm hh}_{ji} \end{pmatrix} = \left(\begin{array}{cc|cc} \rho^{\rm ee}_{i\uparrow j\uparrow} & \rho^{\rm ee}_{i\uparrow j\downarrow} & \rho^{\rm eh}_{i\uparrow j\uparrow} & \rho^{\rm eh}_{i\uparrow j\downarrow} \\ \rho^{\rm ee}_{i\downarrow j\uparrow} & \rho^{\rm ee}_{i\downarrow j\downarrow} & \rho^{\rm eh}_{s\downarrow\uparrow} & \rho^{\rm eh}_{i\downarrow j\downarrow} \\ \hline \rho^{\rm he}_{i\uparrow j\uparrow} & \rho^{\rm he}_{i\uparrow j\downarrow} & \rho^{\rm hh}_{i\uparrow j\uparrow} & \rho^{\rm hh}_{i\downarrow j\uparrow} \\ \rho^{\rm he}_{i\downarrow j\uparrow} & \rho^{\rm he}_{i\downarrow j\downarrow} & \rho^{\rm hh}_{i\uparrow j\downarrow} & \rho^{\rm hh}_{i\downarrow j\downarrow} \end{array}\right)$$

Note the use of the bold to denote that an object possesses a spin structure. Moreover, notice that, since we considered the interaction to be spin-independent, we also have that $V^{i\sigma_i j\sigma_j} \equiv V^{ij}$. In this basis, the self-energies read as

$$\begin{aligned} \Sigma^H_{i\sigma_i j\sigma_j} &= \sum_s V^{is}\left(\rho^{\rm ee}_{s\uparrow s\uparrow} + \rho^{\rm ee}_{s\downarrow s\downarrow}\right)\delta_{ij}\delta_{\sigma_i\sigma_j} \qquad (6) \\ \Sigma^F_{i\sigma_i j\sigma_j} &= -V^{ij}\rho^{\rm ee}_{i\sigma_i j\sigma_j} \text{ and } \Delta_{i\sigma_i j\sigma_j} = V^{ij}\rho^{\rm eh}_{i\sigma_i j\sigma_j} \end{aligned}$$

Notice that the Hartree term is exclusively an onsite due to the δ_{ij}. If we now define a new object related to the Nambu rDM but specifically having the same symmetries as the Nambu Hamiltonian, this being

$$\tilde{\boldsymbol{\rho}}_{ij} = \begin{pmatrix} \boldsymbol{\rho}^{\rm ee}_{ij} & -\boldsymbol{\rho}^{\rm eh}_{ij} \\ -\left(\boldsymbol{\rho}^{\rm eh}_{ji}\right)^* & -\boldsymbol{\rho}^{\rm ee}_{ji} \end{pmatrix}, \qquad (7)$$

we can then write the total self-energy of the system in an elegant and compact form as

$$\mathbf{\Sigma}_{ij} = \frac{1}{2}\sum_s V^{is}{\rm Tr}\left(\left[\tau_z\otimes\sigma_0\right]\tilde{\boldsymbol{\rho}}_{ss}\right)\left[\tau_z\otimes\sigma_0\right]\delta_{ij} - V^{ij}\tilde{\boldsymbol{\rho}}_{ij},$$

with both σ and τ as Pauli matrices but operating in their respectively spin and electron-hole subspaces the Hamiltonian.

Moreover, in the limiting case of a spin-1/2 Hubbard model one considers $V^{ij} = U\delta_{ij}$ such that mean field decoupling gives rise to just the onsite term,

$$\boldsymbol{\Sigma}_{ij} = \frac{1}{2}U\left[\mathrm{Tr}\left(\left[\tau_z \otimes \sigma_0\right]\tilde{\boldsymbol{\rho}}_{ii}\right)\left[\tau_z \otimes \sigma_0\right] - 2\tilde{\boldsymbol{\rho}}_{ij}\right]\delta_{ij}. \quad (8)$$

II. MINIMAL MAJORANA MODEL

First and foremost, prior to a self-consistence discussion, we showcase the foundation of Majorana physics through the study of the minimal model from which Majoranas emerge. Once this groundwork is laid, we hopefully look for similar results in our self-consistent model. The minimal model consists of a finite one-dimensional s-type superconductive (SC) nanowire of static pairing Δ_0 in the presence of an external Zeeman field V_Z whilst accounting for spin-orbit coupling (SOC), being described by the Hamiltonian,

$$\hat{\boldsymbol{H}} = \sum_{\sigma_i\sigma_j} c^\dagger_{\sigma_i}\left(-\eta\partial_x^2 - \mu + \alpha_\mathrm{R} i\sigma_y\partial_x + V_\mathrm{Z}\sigma_z\right)c_{\sigma_j} + \frac{1}{2}(\Delta_0 c^\dagger_\uparrow c^\dagger_\downarrow + \Delta_0 c_\downarrow c_\uparrow). \quad (9)$$

with $\eta = \hbar^2/2m^*$ and m^* the electron effective mass and μ the chemical potential. The SOC being due an externally applied transverse magnetic field of Rashba strength α_R and the Zeeman splitting $V_\mathrm{Z} = g_J\mu_B B_x/2$ due to a in-plane magnetic field B_x, with g_J the Landé $g$yromagnetic factor of the semiconductor and μ_B Bohr's magneton.

Furthermore, within the Nambu tight-binding scheme presented in the previous section, having defined the spinor $\check{\boldsymbol{c}}_i = (c_{i\uparrow}\ c_{i\downarrow}\ c^\dagger_{i\uparrow}\ c^\dagger_{i\downarrow})$, the corresponding Hamiltonian matrix reads as

$$\boldsymbol{H} = (f\delta_{ij} - t\delta_{ij\pm1})\left[\tau_z \otimes \sigma_0\right] + \Delta_0\left[\tau_y \otimes \sigma_y\right]\delta_{ij} \quad (10)$$

$$+ \frac{\alpha_\mathrm{R}}{2a_0}\delta^\pm_{ij\pm1}\left[\tau_z \otimes i\sigma_y\right] + V_\mathrm{Z}\left[\tau_z \otimes \sigma_z\right]\delta_{ij} \quad (11)$$

with $f = 2t - \mu$ the band filling ($\mu = 0$ and $\mu = 2t$ corresponding to half and zero filling respectively), $t = \eta/a_0^2$ the hopping parameter. The Kronecker delta terms δ_{ij} and $\delta^\pm_{ij\pm1}$ denote the on-site terms and the 1st neighbors hoppings, front and back.

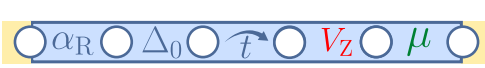

Figure 1. Caption

Before anything else, let us initially consider the case of an *infinity* wire. For starters, we show in Fig.(2) how the SOC influences the energy-momentum band dispersion, plotting in colors the band structure with $\alpha_\mathrm{R} > 0$ and in subtle gray with $\alpha_\mathrm{R} = 0$. Showcased is the shallow and deep electron band (in red) and the respective time-reversed hole's bands $E_{h\pm} = -E_{e\pm}$ (in blue). On the left we set $V_\mathrm{Z} = \Delta_0 = \mu = 0$ to highlight the SOC behavior as it goes from a spinless regime to a helical one as the spin-↑ and spin-↓ bands mitose apart. ★ canting ★ s-wave to p-wave. On the right, we specifically set the value of the Zeeman splitting to it's critical value $V_\mathrm{Z}^c = \sqrt{\mu^2 + \Delta_0^2}$ (having nonzero Δ_0 and μ), where the band gap at $k = 0$ closes up. It's the closing and reopening of the gap at $k = 0$ that signals the trivial-topological phase transition. This type of phase transition is said to not be symmetry protected system, meaning that it isn't possible to go from a topological state to a trivial one without going through a closing of gap. Furthermore, see that as one accounts for SOC effects there is an additional minigap that opens ups on the outer wings. Thus, for values $V_\mathrm{Z} < V_\mathrm{Z}^c$ the nanowire is in it's trivial phase, having a superconducting gap and being no longer spinless, while for values $V_\mathrm{Z}^c > V_C$ the nanowire is in it's topological phase, opening up a minigap due to the combined affects of Zeeman and SOC, eliminating the possibility of high-momentum gapless excitations, leaving only the chiral states near $k = 0$ as low energy excitation.

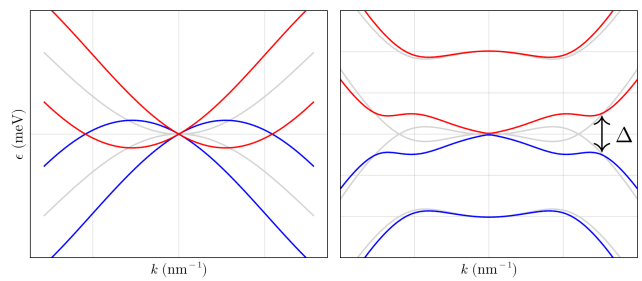

Figure 2. Finite nanowire energy-momentum dispersion $E(k)$ for different values of chemical potential μ, Zeeman field V_Z, static pairing potential Δ_0. The Rashba interaction strength is set to $\alpha_\mathrm{R} = 1$ unless otherwise specified and the effective mass m^* is set such that $\eta = 1$.

Furthermore, in Fig.(3) we show that this trivial-topological phase transition, controlled by the tuning of the chemical potential μ and Zeeman field V_Z whilst fixing the static pairing Δ_0 and accounting for a strong SOC, is denoted by the emergence of two Majorana zero energy states which are topological protected through the minigap. ★ Majoranas These Majorana's wavefunctions rapidly decay from the edges out into the center, however, for small enough nanowire lengths the wavefunctions can actually overlay hybridizing into fermions of oscillatory energy around zero (we will not focus on such cases though).

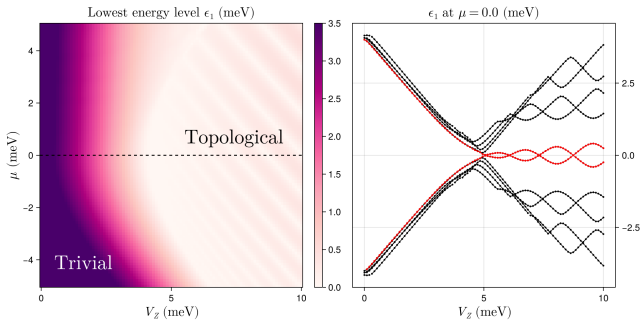

Figure 3. **(right)** Trivial-topological phase transition diagram traced by the lowest energy state, and **(left)** the remaining energy spectrum for $\mu = 0$ (dashed black line), for a SC nanowire of length $L = 500$nm of static pairing $\Delta_0 = 4$meV (toy-parameter, not realistic). The spacing was set to $a_0 = 10$nm and the effective mass to $m_0 = 0.023$meV, corresponding to InAs.

III. SELF-CONSISTENT NANOWIRES

As a step up to the minimal Majorana model, in place of statically fixing the pairing Δ_0, we instead solve for it self-consistent. Thus, in place of the superconductive proximity effect term in Eq.(10), we use instead the on-site Hubbard mean-field self-energy Σ_{ij} as derived in Eq.(**??**). This model is pictorially depicted in Fig.(4).

Although looking for Majoranas physics, we first focus on the infinite length nanowire case because it is a lot less numerically expensive and because it lays out the groundwork analysis without muddying it with finite length effects, which unnecessarily complicates the analysis.

Figure 4. Models

We self-consistently obtain the rDM through the Hamiltonian's Green's function $\boldsymbol{G}$ for the finite case through a spectrum diagonalization, so basically $\rho = -i(\epsilon + i\eta - H)^{-1}$, and for the infinite case through a generalized Schur's factorization relying on the numerical integration over Bloch phase $\rho_{dn} = \frac{1}{2\pi}\int_{-\pi}^{+\pi} d\phi f(H(\phi) - \mu I)e^{idn\phi}$ where the Fermi function f of the Bloch Hamiltonian is computed via exact diagonalization at each Bloch phase ϕ. ★ finite temperature? As a side-note, it it important that one forcefully imposes on every iteration of the self-consistency the BdG particle-hole symmetries as expressed rightly after Eq.(2) because numerical errors will progressively drive the result to break those symmetries, even though they are self-imposed in the original Hamiltonian.

In Fig.(5), we first study the phase diagrams of the superconductive and magnetic correlation, corresponding respectively to $\Delta = \rho^{\text{eh}}_{i\uparrow j\downarrow}$ and $M = (\rho^{\text{ee}}_{i\uparrow j\uparrow} - \rho^{\text{ee}}_{i\downarrow j\downarrow})/2$, as one tunes the on-site Hubbard potential U versus the the chemical potential μ and as one tunes the chemical potential μ versus the Zeeman field strength. The analysis boils down to the ongoing competition between those two quantities, Δ and M. Looking at Fig.(5)(left) we see that, for a given reasonable μ and V_Z, if $U < 0$ then only Δ is non-zero and the system is in it's superconductive phase and if $U > 0$ then only M in non-zero and the system is in it's ferromagnetic phase. Note that while $U < 0$ is always a prerequisite for superconductive, it alone does not guarantee it. Firstly because a strong enough external magnetic field B would still magnetize the system and secondly because BCS theory tells us that $\Delta_{\text{BCS}} \approx \omega_{\text{D}} e^{(-U n_{\text{F}} a_0)^{-1}}$ with n_{F} being the density of states at the Fermi level which in 1D decreases as μ increases, thus meaning that also for a sufficient large μ superconducting dies out. On another matter, in Fig.(5)(right) we show how the superconductive correlation behaves during the topological phase transitions. In this case the phase transition (traced in black) is found through the abrupt change in value of Kitaev's "Majorana number" $\mathcal{M}$, being either $\mathcal{M} = +1$ in the trivial case or $\mathcal{M} = -1$ in the topological case (applicable only in the case of open boundary conditions) [?]. For general non-interacting systems, the fermionic parity P of the ground state of a Hamiltonian H can be shown to take the form $P(H) = \text{sgn}(\text{Pf}(A))$ with the Pfaffian Pf a function of a skew-symmetric matrix A such that $(\text{Pf}(A))^2 = \det(A)$. For this reason, $\mathcal{M}$ is also referred to as the Pffafian $\mathbb{Z}_2$ invariant.

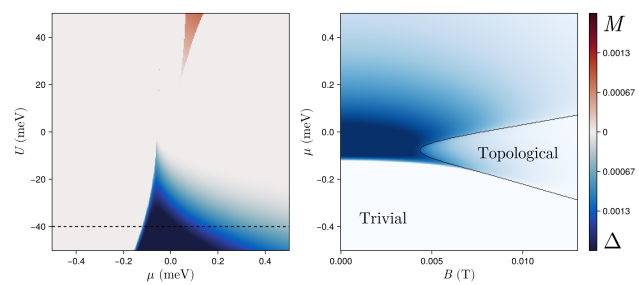

Figure 5. Phase Diagrams

★ correct window of value, explaining it in terms of the minigap and magnetization limits.

In the heart of an ancient, dense forest, where the trees whispered secrets of old and the breeze carried tales of forgotten times, stood the Temple of the Crimson Canopy. Within its walls lived a community of monkeys, led by an alpha acolyte named Jungly Pirulo, draped in a majestic red garment that signified his authority and wisdom.

Jungly Pirulo was not just any monkey; he was revered by his kin for his exceptional intelligence, strength, and unwavering devotion to the ancient ways of their temple. From a young age, he had shown promise, mastering the teachings of the elders with unmatched zeal. His red

garment, adorned with intricate patterns symbolizing the unity of the forest and the spirits that dwelled within, marked him as the chosen leader of his clan.

Under Pirulo's guidance, the monkeys of the Crimson Canopy thrived. They spent their days in harmony with the rhythms of the forest, tending to the temple grounds, and meditating beneath the sheltering canopy of ancient trees. But their tranquil existence was soon disrupted by an unforeseen threat.

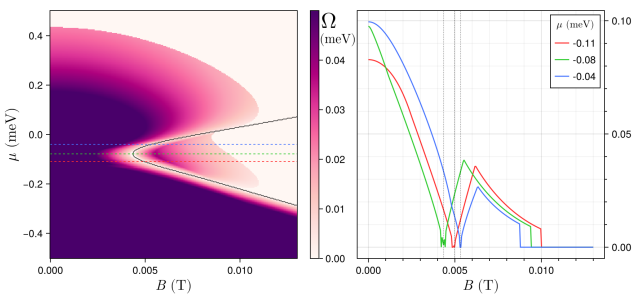

Figure 6. Phase Diagrams capped colormap

A group of humans, driven by greed and blinded by their desire for riches, ventured deep into the forest, intent on exploiting its resources. They saw the Temple of the Crimson Canopy as nothing more than a hindrance to their ambitions and set out to claim the land for themselves.

Pirulo sensed the danger looming on the horizon and knew that the time had come to defend his home. Gathering his fellow acolytes, he devised a plan to protect the temple and preserve the sanctity of their forest sanctuary.

As the humans encroached upon their territory, armed with axes and machines of destruction, Pirulo and his followers launched a daring counterattack. With agility and cunning, they outmaneuvered their adversaries, using the dense foliage to their advantage and striking with the precision of seasoned warriors.

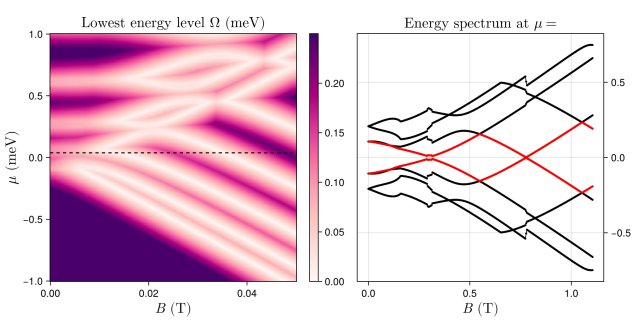

Figure 7. Phase Diagrams

IV. SELF-CONSISTENT JOSEPHSON JUNCTION

Consider a Majorana 1D nanowire along the $\hat{\mathbf{z}}$ direction as described in Fig.(??)(a).

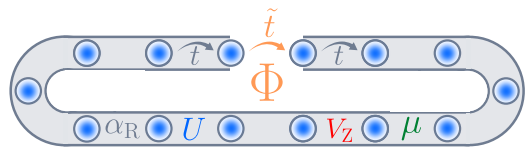

Figure 8. Model

Consider now that the edges $i = 1, N$ of the nanowire connect to one another as a weak link, forming a ring in the plane $x\mathcal{O}y$, as shown in Fig.(??)(c). We then apply a magnetic field B_z through the middle of the ring such that the kinetic term of the Hamiltonian is modified as $(-i\hbar\boldsymbol{\nabla} + e\mathbf{A})^2/2m_0$, with $-e$ the electron's charge and $\mathbf{A}$ the magnetic vector potential. The vector potential is constrained by the magnetic flux Φ through the ring as $\oint d\mathbf{r}\cdot\mathbf{A}(\mathbf{r}) = \Phi$. According to Peierl's substitutions, assuming that the vector potential changes slowly and is approximately uniform at the Wannier lattice scale, the modified hopping $\tilde{t}$ in quasi-equilibrium stays the same apart from the phase factor picked up, reading as

$$\tilde{t}_{ij} = t_{ij}e^{-i\frac{e}{\hbar}\phi_{ij}}, \text{ with } \phi_{ij} = e^{-i\frac{e}{\hbar}\int_{\mathbf{r}_i}^{\mathbf{r}_j} d\mathbf{r}'\cdot\mathbf{A}(\mathbf{r}')}$$

Furthermore, one can then chose a gauge $\mathbf{A}(\mathbf{r}) \to \mathbf{A}(\mathbf{r}') + \boldsymbol{\nabla}\boldsymbol{\Lambda}(\mathbf{r})$ where the vector potential vanishes everywhere except in the weak link, so $\phi_{ij} = \phi\delta_{i1}\delta_{jN}$. The weak link hopping phase reads as $\phi = n\Phi/\Phi_0$ with $\Phi_0 = h/2e$ the fluxoid (superconducting flux quanta) with $2e$ the charge of the Cooper pair and $\Phi = \pi R^2 B_z$ with R the ring radius.

According to the Meissner effect, the screening currents flowing in the London's penetration depth λ_L outer shell create a magnetization which perfectly compensates the applied field, thus resulting in $\mathbf{B} = 0$ within the bulk of the superconductor. However, note that the value of the flux quantum is equal to Φ_0 is only true if the path around the hole lays in the bulk, several λ_L away from the surface. Since we are considering a 1D nanowire this condition is then not satisfied.

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